DELCODE PACC5 / ADAS-Cog – Documentation

The PACC5 and ADAS-Cog composite scores exported with the April 2022 DELCODE release were calculated from quality checked and cleaned neuropsychological data. That is, cases with impossible or highly implausible data entries were excluded and identifiable data entry mistakes were rectified prior to the calculation of the exported variables.

When publishing data analyses using either of the scores, please include descriptions similar to the following paragraphs in your methods section, if space allows.

1. ADAS-Cog-11 [AD11SUM]

The classic ADAS-Cog scale according to Rosen et al. (1984) was calculated from 11 tasks assessing the cognitive domains of memory, language, orientation, as well as constructional and ideational praxis. Scores range from 0 to 70, with higher values indicating greater impairment.

2. ADAS-Cog-13 [AD13SUM]

The ADAS-Cog-13 scale proposed by Mohs et al. (1997) was calculated from 13 tasks assessing the cognitive domains of memory, language, orientation, attention and concentration, as well as constructional and ideational praxis. It includes all ADAS-Cog-11 items with the addition of a delayed word recall and number cancellation task. The scale ranges from 0 to 85, with higher values indicating greater impairment.

3. PACC5 [pacc5]

The preclinical Alzheimer´s cognitive composite (PACC5) proposed by Papp et al. (2017) was calculated as the averaged z-standardised performance in five cognitive measures: the MMSE, the summed Free and Cued Selective Reminding Test (FCSRT) free and total recall, the Wechsler Memory Scale – Fourth Edition (WMS-IV) Logical Memory Story B delayed recall, the Symbol-Digit Modalities Test (SDMT), and the sum of two category fluency tasks. Z-scores were derived from the baseline means and standard deviations of cognitively unimpaired DELCODE participants and then averaged across the five measures.

References

Mohs, R. C., Knopman, D., Petersen, R. C., Ferris, S. H., Ernesto, C., Grundman, M., . . . Thai, L. J. (1997). Development of cognitive instruments for use in clinical trials of antidementia drugs: Additions to the Alzheimer's Disease Assessment Scale that broaden its scope. *Alzheimer Disease & Associated Disorders*, *11*, 13–21.

Papp, K. V., Rentz, D. M., Orlovsky, I., Sperling, R. A., & Mormino, E. C. (2017). Optimizing the preclinical Alzheimer’s cognitive composite with semantic processing: The PACC5. *Alzheimer’s & Dementia*, *3*(4), 668–677.

Rosen, W. G., Mohs, R. C., & Davis, K. L. (1984). A new rating scale for Alzheimer’s disease. *The American Journal of Psychiatry*, *141*(11), 1356–1364.

Appendix – PACC5 SPSS syntax

***z-scores from mean baseline performance in MMSE, WMS-R Logical Memory Delayed Recall, Digit-Symbol-Coding-Test (SDMT number of correct answers), FCSRT Free Recall and Total Recall, Categorial Fluency Animal and Food
***z-scores: M and SD from baseline of cognitively normal patients: healthy controls, AD patients' relatives, SCD
***variables: mmstot, wmslm2b, sdmctot, fcsrt96 = (fcstot + fcsfree), cat_sum (vfatot (animals), cervfg (food))

```
***compute composite scores of FCSRT and Categorial Fluency
COMPUTE fcsrt96 = (fcstot + fcsfree).
COMPUTE cat_sum = (vfatot + cervfg).
EXECUTE.

***filter for cognitively normal participants (prmdiag: healthy controls (0), AD patients' relatives (100), SCD (1))
USE ALL.
COMPUTE filter_$ = (visnam_num = 0) AND ((prmdiag = 0) OR (prmdiag = 1) OR (prmdiag = 100)).
VARIABLE LABELS filter_$ 'visnam_num = 0 prmdiag = 0 prmdiag = 1 prmdiag = 100 (FILTER)'.
VALUE LABELS filter_$ 0 'Not Selected' 1 'Selected'.
FORMATS filter_$ (f1.0).
FILTER BY filter_$.
EXECUTE.

DESCRIPTIVES VARIABLES=mmstot wmslm2b sdmctot fcsrt96 cat_sum
/STATISTICS=MEAN STDDEV.
```

	N	Mean	Standard Deviation
MMSE Total	763	29.32	0.97
wmslm2b	761	13.64	3.86
SDMT: Number correct responses	760	47.66	9.72
FCSRT TR + FR	759	78.81	7.11
verbal fluency animals + food	761	48.63	10.42

```
FILTER OFF.
USE ALL.
EXECUTE.

***compute z-scores from baseline performances
COMPUTE z_mmstot = (mmstot - 29.32)/0.965.
EXECUTE.

COMPUTE z_wmslm2b = (wmslm2b - 13.64)/3.863.
EXECUTE.

COMPUTE z_sdmctot = (sdmctot - 47.66)/9.715.
EXECUTE.

COMPUTE z_fcsrt96 = (fcsrt96 - 78.8076)/7.11199.
EXECUTE.

COMPUTE z_cat_sum = (cat_sum - 48.6334)/10.42130.
EXECUTE.

***compute PACC5 score
COMPUTE pacc5 = (z_mmstot + z_wmslm2b + z_sdmctot + z_fcsrt96 + z_cat_sum)/5.
EXECUTE.
```